INTELLIGENT ORDER ASSISTANT

1. Background

With the rapid growth of e-commerce and digital services, customers today expect quick, accurate, and always-available support for their queries, especially related to orders, tracking, and service assistance. Traditional customer support systems often rely on human agents, which can lead to delays, high workload, and inconsistent responses during peak usage times.

To overcome these limitations, AI-powered chatbot systems have emerged as an effective solution. These systems use Artificial Intelligence and Natural Language Processing to understand user queries and provide instant responses. **Sevabot AI – Intelligent Order Assistance Bot** is developed to improve customer experience by offering real-time, automated, and intelligent support for order-related tasks through a simple and accessible web interface.

2. Requirement Overview

The system is designed to fulfill the following requirements:

- Provide real-time AI-based chat support to users
- Assist users with order-related queries (placement, tracking, status)
- Maintain conversational context for better understanding
- Offer a simple and user-friendly web interface
- Ensure fast and accurate responses using AI models
- Support scalable deployment for multiple users
- Integrate with backend APIs for dynamic responses

3. Solution Approach

The solution is built using a **full-stack AI-powered architecture**:

1. User interacts with a chatbot UI on the website
2. Messages are sent to backend via REST API

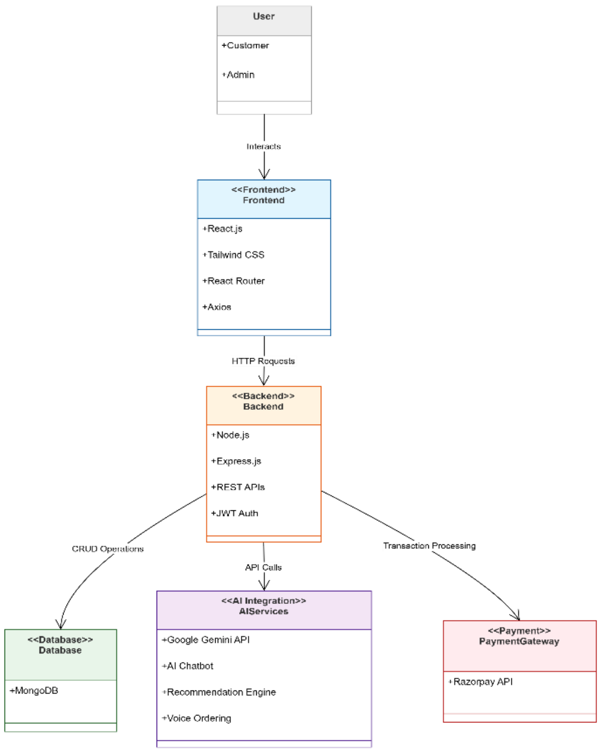
3. Backend processes the request and communicates with AI model (OpenAI/Gemini/custom LLM)
4. AI generates contextual response
5. Response is sent back to frontend and displayed to user

This approach ensures real-time communication and scalable AI integration.

4. Solution Architecture

The architecture of the system follows a client-server model using the MERN stack and AI integration.

Architecture Flow



The frontend communicates with backend APIs, while the backend handles database operations, AI processing, authentication, and payment services.

5. Technical Details

Frontend Technologies

- React.js / HTML / CSS
- Provides interactive chatbot interface

- Handles user input and displays AI responses
- Responsive design for mobile and desktop

Backend

- Node.js + Express.js
- Handles API requests and routing
- Manages communication between frontend and AI service

AI Integration

- OpenAI API / Google Gemini API
- Processes natural language queries
- Generates intelligent and contextual responses

Deployment

- Google Cloud Run
- Provides scalable, container-based deployment
- Ensures high availability and performance

Why these technologies?

- React ensures fast UI rendering
- Node.js provides efficient non-blocking backend
- AI APIs enable intelligent responses without training models
- Cloud Run ensures easy scaling and deployment

6. Benefits of the Solution

The Intelligent Order Assistant provides several advantages over traditional e-commerce systems:

- Fast and real-time responses
- Automated customer support reduces manual workload
- Scalable system for growing users
- Intelligent and context-aware conversations

- Reduces operational cost for businesses
- Accessible from any device via web

The system creates a modern, intelligent, and user-friendly shopping platform.

7. Alternate Approach

An alternative approach for developing the system could involve using different technologies and architectures.

Possible alternatives include:

- Using Firebase instead of MongoDB for real-time database management
- Developing the application as a mobile app using Flutter or React Native
- Using Python Django instead of Node.js for backend development
- Implementing rule-based chatbot systems instead of AI-based conversational assistants
- Using SQL databases such as MySQL instead of MongoDB

However, the MERN stack was selected because it provides:

- Better JavaScript ecosystem integration
- Faster development
- Scalability
- Flexible NoSQL database structure
- Efficient frontend-backend communication

8. Assumptions

The following assumptions were considered during system development:

- Users have internet access to use the chatbot
- AI API (OpenAI/Gemini) is available and functional
- Backend server is properly hosted and running
- Users interact only through web interface
- Order data (if integrated) is assumed to be available via API or database
- System is designed for English language interactions

9. Screenshots of App + Power BI

Power BI Dashboard:

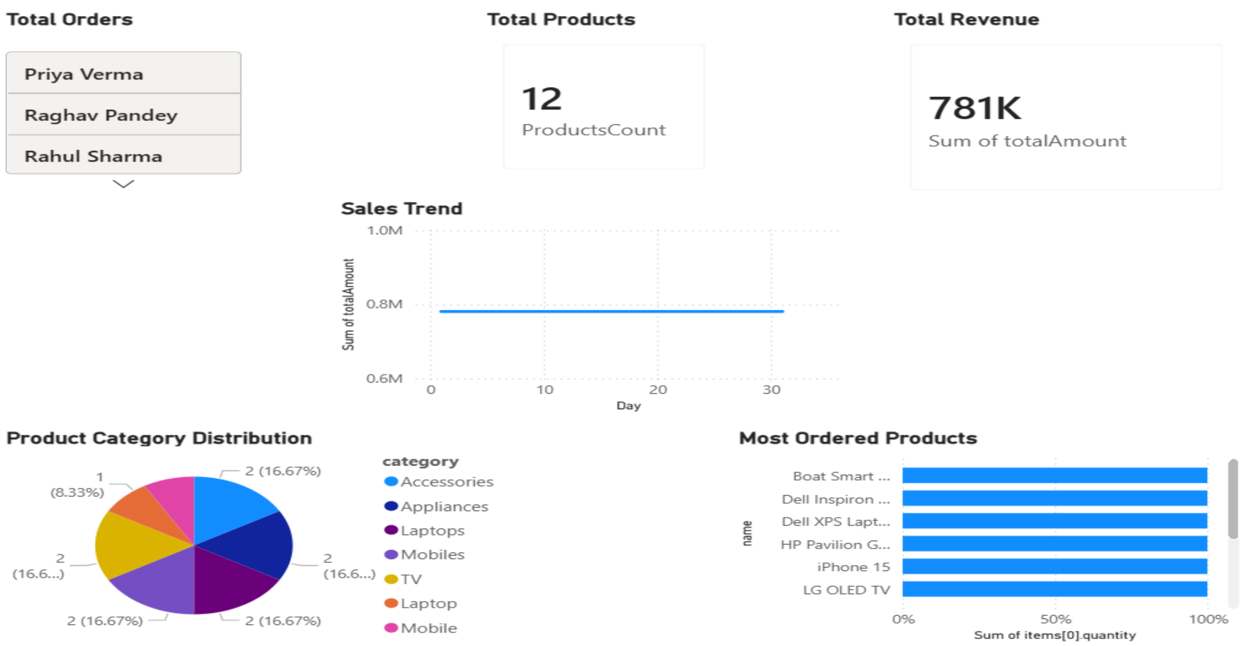
The project includes a Power BI analytics dashboard for visualizing ecommerce data and generating business insights. The dashboard was developed using Microsoft Power BI Desktop by importing exported MongoDB collections in CSV format.

The dashboard provides:

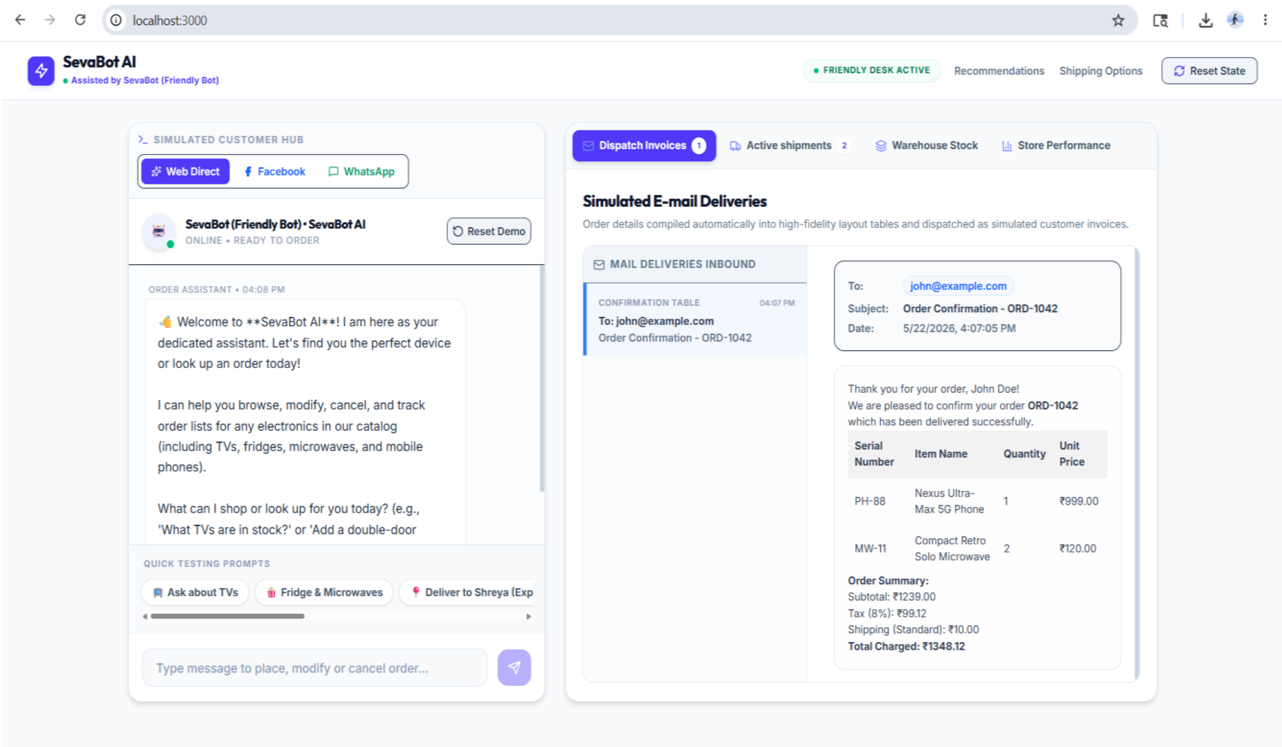
- Total Orders Analysis
- Total Revenue Analysis
- Total Products Count
- Product Category Distribution
- Most Ordered Products Visualization
- Sales Trend Analysis

Different visualizations such as cards, pie charts, bar charts, and line charts were used to present the data in an interactive and user-friendly manner.

The dashboard helps administrators monitor product performance, customer purchasing behavior, and overall sales trends effectively.



Screenshot:


localhost:3000
SevaBot AI
Assisted by SevaBot (Friendly Bot)
FRIENDLY DESK ACTIVE
Recommendations
Shipping Options
Reset State
SIMULATED CUSTOMER HUB
Web Direct
Facebook
WhatsApp
SevaBot (Friendly Bot) • SevaBot AI
ONLINE • READY TO ORDER
Reset Demo
ORDER ASSISTANT • 04:08 PM
Welcome to **SevaBot AI**! I am here as your dedicated assistant. Let's find you the perfect device or look up an order today!
I can help you browse, modify, cancel, and track order lists for any electronics in our catalog (including TVs, fridges, microwaves, and mobile phones).
What can I shop or look up for you today? (e.g., 'What TVs are in stock?' or 'Add a double-door
QUICK TESTING PROMPTS
Ask about TVs
Fridge & Microwaves
Deliver to Shreya (Exp
Type message to place, modify or cancel order...
Dispatch Invoices 1
Active shipments 2
Warehouse Stock
Store Performance
Simulated E-mail Deliveries
Order details compiled automatically into high-fidelity layout tables and dispatched as simulated customer invoices.
MAIL DELIVERIES INBOUND
CONFIRMATION TABLE
04:07 PM
To: john@example.com
Order Confirmation - ORD-1042
To: john@example.com
Subject: Order Confirmation - ORD-1042
Date: 5/22/2026, 4:07:05 PM
Thank you for your order, John Doe!
We are pleased to confirm your order ORD-1042 which has been delivered successfully.
Serial Number
Item Name
Quantity
Unit Price
PH-88
Nexus Ultra-Max 5G Phone
1
₹999.00
MW-11
Compact Retro Solo Microwave
2
₹120.00
Order Summary:
Subtotal: ₹1239.00
Tax (8%): ₹99.12
Shipping (Standard): ₹10.00
Total Charged: ₹1348.12